

Learner-State-Aware Generative User Interfaces: A Closed-Loop Control Architecture for Adaptive Learning Environments

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Abstract—Intelligent tutoring systems estimate what a learner knows but present a fixed interface, while generative user interfaces (GenUI) synthesize interfaces at runtime without any model of the learner. We propose a closed-loop architecture that couples the two. Bayesian Knowledge Tracing estimates mastery of a knowledge component; a discrete-time PID governor with anti-windup clamping and a filtered derivative converts the mastery deficit into a structural complexity budget; a fast policy selects discrete scaffolding affordances; and a foundation model emits a declarative JSON component tree that must satisfy the budget, with a deterministic fallback when schema or budget checks fail. A computable metric of interface complexity closes the loop between the budget and the rendered tree. We evaluate the architecture in silico on 1,000 synthetic learner trajectories drawn from three learner archetypes and replayed through four experimental arms. We report constraint adherence, interface jitter, recovery after error streaks and schema compliance **[TBD: headline numbers]**. We further show that standard BKT saturates after long success runs, so that a subsequent streak of errors leaves the interface unchanged, and evaluate forgetting and a CUSUM lapse detector as remedies. All claims concern algorithmic behavior; no learning gains are claimed.

Index Terms—generative user interfaces, Bayesian knowledge tracing, PID control, cognitive load, adaptive learning, simulation

I. INTRODUCTION

Intelligent tutoring systems (ITS) have long modeled learners precisely. Knowledge tracing maintains a probability that a learner has mastered each knowledge component [1], [2], and tutors built on it have been deployed at scale in mathematics [3], [4]. The interface of such systems, however, is authored in advance: the learner model selects the next problem, not the form in which the problem is presented.

Generative user interfaces take the opposite position. A foundation model produces a task-specific interface for each request [5], including interactive courseware [6]. These systems are open loop: the model does not observe the learner’s state, and nothing bounds how complex the generated interface becomes.

Cognitive load theory explains why this matters. The same material can help a novice and hinder an expert (the expertise reversal effect) [7], [8], and guidance should fade as competence grows [9], [10]. An interface that adapts to the learner therefore needs both a learner model and a mechanism that holds the interface to a complexity appropriate to that

model, without the abrupt layout changes that make adaptive interfaces hard to use [11], [12].

This paper makes three contributions:

- 1) A closed-loop architecture in which a BKT mastery estimate drives a PID governor whose output is a structural complexity budget that a generative model must satisfy, with schema validation and a deterministic fallback.
- 2) A computable structural complexity metric over a declarative component tree for linear-equation tasks, built from features linked in prior work to cognitive load and visual complexity.
- 3) An in-silico benchmark of 1,000 synthetic learners across four arms and ablations, including an analysis of BKT saturation after success runs and two remedies.

II. RELATED WORK

A. Learner modeling

Bayesian Knowledge Tracing (BKT) models each knowledge component as a two-state hidden Markov model with prior, learning, guess and slip parameters [1]. Extensions individualize the prior [13] and the learning rate [14], add forgetting [15], [16], or contextualize guess and slip [17]. Its identifiability and degeneracy have been analyzed in detail [18]–[20]. Logistic models such as PFA [21], [22] and deep models [23] are alternatives; BKT remains competitive when extended [16] and is reviewed in [24], [25]. We use BKT because its state is a calibrated probability that can serve directly as a control variable.

B. Adaptive and generative interfaces

Adaptive hypermedia [26] and model-based interface generation [27] adapt presentation to users, but adaptation has costs: unpredictable changes reduce usability [11], [28], [29]. Recent GenUI systems let a large language model produce the interface itself [5], [6]. To our knowledge, none of these systems bounds the generated interface by a learner model.

C. Cognitive load and scaffolding

Cognitive load theory [30]–[32] attributes intrinsic load to element interactivity [33]. Worked examples reduce load for algebra novices [34] and should fade into problem solving [9], [10]; concrete representations should fade into symbolic ones [35]. For linear equations, the balance model supports

early understanding but breaks down with negative quantities [36], [37], and learners hold persistent misconceptions about equality and operating on the unknown [38]–[41].

D. Control and constrained generation

PID control with anti-windup and derivative filtering is standard practice [42]–[44], and feedback control has been applied to computing systems [45] and to difficulty adjustment in games [46]. Constrained decoding guarantees well-formed model output [47]–[50], though schema-constrained output is not perfectly reliable in practice [51] and format restrictions can affect output quality [52].

III. ARCHITECTURE

The loop has four layers (Fig. [TBD: architecture figure]): a psychometric estimator, a complexity governor, a fast policy, and a generative plant whose output passes a validation gate before a deterministic client renders it.

A. Mastery estimation

After response $r_n \in \{0, 1\}$ the mastery estimate is updated by Bayes' rule and the learning transition [1]:

$$P(L_n | r_n=1) = \frac{P(L_n)(1 - P(S))}{P(L_n)(1 - P(S)) + (1 - P(L_n))P(G)}, \quad (1)$$

$$P(L_n | r_n=0) = \frac{P(L_n)P(S)}{P(L_n)P(S) + (1 - P(L_n))(1 - P(G))}, \quad (2)$$

$$P(L_{n+1}) = P(L_n | r_n)(1 - P(F)) + (1 - P(L_n | r_n))P(T), \quad (3)$$

where $P(F) = 0$ gives standard BKT and $P(F) > 0$ adds forgetting [16]. Default parameters are $P(L_0) = 0.15$, $P(T) = 0.12$, $P(G) = 0.20$, $P(S) = 0.08$. They are design values, not fitted; they satisfy the bounds $P(G) < 0.3$, $P(S) < 0.1$ [1] and the non-degeneracy conditions $P(G) + P(S) < 1$, $P(T) < 1 - P(S)/(1 - P(G))$ [19].

B. Complexity governor

The setpoint is the conventional mastery threshold $P^* = 0.95$ [1], [53]. The tracking error is normalized on each side of the setpoint so that deficit and surplus have equal authority:

$$e_n = \begin{cases} (P^* - P(L_n))/P^* & P(L_n) \leq P^* \\ (P^* - P(L_n))/(1 - P^*) & P(L_n) > P^* \end{cases}, \quad e_n \in [-1, 1]. \quad (4)$$

The control effort is

$$u_n = K_p e_n + K_i S_n + K_d D_n, \quad (5)$$

with the clamped integral $S_n = \text{clamp}(S_{n-1} + e_n, -S_{\max}, S_{\max})$ for anti-windup [44] and the first-order filtered derivative $D_n = \gamma(e_n - e_{n-1}) + (1 - \gamma)D_{n-1}$ [42]. An inverted logistic maps effort to the complexity budget,

$$M_I^* = (1 + e^{ku_n})^{-1}, \quad (6)$$

which is quantized into four levels $\ell_n = \min(4, 1 + \lfloor 4M_I^* \rfloor)$. A hysteresis deadband Δ_h commits a new budget only when it differs from the last committed budget by at least Δ_h .

Gain selection: Classical tuning rules [54]–[57] assume an identified plant response. In our benchmark the interface does not alter the simulated learner's responses (Section IV), so there is no plant response to identify. Instead, K_p , K_i , K_d , γ , $S_{\max}$ and Δ_h were selected by minimizing an ITAE-type tracking cost [58] plus a jitter penalty on a separate tuning cohort; all reported results use a disjoint test cohort. Selected values: [TBD: tuned gains from results/]. The logistic slope k is chosen so that the steady-state effort range maps to approximately $[0.05, 0.95]$ [TBD: re-derive k for $P^* = 0.95$].

C. Fast policy

A fast policy maps $(P(L_n), M_I^*, \ell_n, \text{recent errors})$ to discrete affordances: scaffolding mode (worked example, guided scaffold, open symbolic), whether to show the balance-scale manipulative, whether hints are enabled, and an instructional density limit. The intended implementation is an open-weight decision model [59]; the benchmark uses a deterministic rule-based stand-in with the same interface, so no latency claims about that model are made. The split between a fast decider and a slow generator follows the System 1 / System 2 analogy [60].

D. Generative plant and validation gate

The plant ([TBD: Gemini model and configuration]) receives the budget, the policy decision and the equation and emits a JSON component tree. Nodes are drawn from a closed set of six components: CONCRETEBALANCESCALE, WORKEDSOLUTIONSTEP, SCAFFOLDEDOPERATIONPAD, SYMBOLICEQUATIONWORKSPACE, STEPPEDHINTACCORDION and CARTESIANVERIFICATIONPLANE. The client renders these components directly and never executes generated code. Each emission is parsed against the schema [61]; if parsing fails, or if $|M_I - M_I^*| > \varepsilon$ with $\varepsilon = 0.05$, a precompiled fallback tree nearest to the budget is rendered instead.

E. Structural complexity metric

For a tree with ρ interactive elements, abstraction level α and nesting depth δ ,

$$M_I = w_1 \frac{\rho - 1}{7} + w_2 \frac{\alpha - 1}{3} + w_3 \frac{\delta - 1}{3}, \quad \rho \in [1, 8], \alpha, \delta \in [1, 4]. \quad (7)$$

Each feature has a basis in prior work: element interactivity and choice count raise load [33], [62], [63] and display density is a classic complexity measure [64]; the abstraction ladder from balance scale to operation pad to symbolic workspace follows concreteness fading [35]; and deeper structures cost users more [65], [66]. Placing the Cartesian plane at the top of the ladder is our design choice. With no data to fit weights, we use equal weights $w_i = 1/3$, which are robust in that setting [67], and test sensitivity to the weights (Section V). M_I is a structural proxy; it has not been validated against measured cognitive load.

TABLE I
LEARNER ARCHETYPES (UNIFORM PARAMETER RANGES).

Archetype	$P(L_0)$	$P(T)$	$P(G)$	$P(S)$
Struggling novice	.02–.08	.03–.07	.10–.20	.20–.30
Inconsistent guesser	.05–.20	.10–.20	.15–.40	.08–.30
Fast master	.15–.30	.30–.40	.15–.25	.01–.03

IV. METHOD

A. Synthetic learners

Simulated learners are an established way to test instructional designs before human studies [68], [69], and BKT-generated data have been used to evaluate mastery assessment [70], [71]. Each learner is a two-state HMM whose parameters are drawn uniformly from archetype ranges (Table I). The ranges for the novice and guesser archetypes exceed the tighter bounds of [1] by design to represent noisy learners, while respecting the non-degeneracy conditions of [19]. The cohort has 1,000 learners (300/400/300) with 40 items each. Responses are sampled once per learner and replayed unchanged through every arm (paired design). The interface therefore does not influence the simulated responses: the benchmark measures controller behavior, not learning.

B. Arms

(1) *Static*: a fixed level-2 interface. (2) *Unconstrained GenUI*: the plant chooses complexity from a noisy reading of mastery with no budget check. (3) *Rule-based*: step up after three consecutive correct responses and down after two consecutive errors, an up-down rule in the spirit of [72], [73]. (4) *Closed loop*: the full architecture. Ablations remove the derivative filter, remove anti-windup, and add a one-level slew limit.

C. Plant model in the benchmark

[TBD: If real Gemini rates are measured: describe them here. Otherwise:] The benchmark replaces the foundation model with a surrogate that compiles the closest admissible tree and injects faults: with probability [TBD: 0.15] it drifts off budget, and with probability [TBD: 0.03] it emits malformed output (truncated JSON, unknown component, out-of-range field, depth overflow, injected field). These rates are assumptions, not measurements.

D. Metrics

Constraint error $\Delta M = |M_I - M_I^*|$; schema compliance (valid trees over attempts); interface jitter $J = \frac{1}{N-1} \sum_n |M_I(n) - M_I(n-1)|$; recovery time after a five-error streak; and fallback rate. Jitter operationalizes the stability concern of [11]; the formula is ours.

V. RESULTS

[TBD: All numbers in this section come from the confirmed final benchmark run (commit, seed and CSV columns to be supplied by the engine).]

TABLE II
MAIN RESULTS ON THE TEST COHORT (MEAN $\pm$ 95% CI).

Arm	ΔM	Jitter J	Schema ok	Fallback
Static	[TBD:]	[TBD:]	[TBD:]	[TBD:]
Unconstrained GenUI	[TBD:]	[TBD:]	[TBD:]	[TBD:]
Rule-based	[TBD:]	[TBD:]	[TBD:]	[TBD:]
Closed loop	[TBD:]	[TBD:]	[TBD:]	[TBD:]

A. Ablations

[TBD: derivative filter, anti-windup, slew limit].

B. Sensitivity

[TBD: gains at $\pm 50\%$; metric weights: equal, 0.40/0.35/0.25, and each term dropped].

C. BKT saturation and lapse handling

Under correct responses the BKT estimate converges to a stable fixed point at one [19]. With the default parameters the estimate exceeds $1 - 10^{-6}$ after ten correct responses and five subsequent errors lower it only to about 0.98; after roughly forty correct responses it equals 1.0 in double precision and no further evidence can lower it. The interface therefore stays at its highest level through a streak of errors. We compare two remedies: forgetting, $P(F) > 0$ [15], [16], and a CUSUM detector [74], [75] on the log-likelihood ratio between “not known” and the tracker’s prediction, which resets the estimate when an alarm fires. [TBD: results on the forgetting cohort]. Weighting recent responses more heavily [76] is a third option we leave for future work.

VI. DISCUSSION AND LIMITATIONS

The benchmark tests whether the controller does what it is designed to do; it does not test whether learners benefit. Because simulated responses do not depend on the interface, the loop is closed only through the mastery estimate, not through the learner. M_I is a structural proxy that has not been validated against measured load, for which instruments exist [77]–[79]. BKT parameters are design values rather than fits to data such as ASSISTments [80]. Long error streaks may reflect misconceptions rather than slips [41], or a learner who will not reach mastery [81]; the controller does not distinguish these. [TBD: plant: surrogate vs. measured Gemini rates].

VII. CONCLUSION

[TBD: one paragraph after results are final].

REFERENCES

- [1] A. T. Corbett and J. R. Anderson, “Knowledge tracing: Modeling the acquisition of procedural knowledge,” *User Modeling and User-Adapted Interaction*, vol. 4, no. 4, pp. 253–278, 1995.
- [2] K. R. Koedinger, A. T. Corbett, and C. Perfetti, “The knowledge-learning-instruction framework: Bridging the science-practice chasm to enhance robust student learning,” *Cognitive Science*, vol. 36, no. 5, pp. 757–798, 2012.
- [3] S. Ritter, J. R. Anderson, K. R. Koedinger, and A. Corbett, “Cognitive tutor: Applied research in mathematics education,” *Psychonomic Bulletin & Review*, vol. 14, no. 2, pp. 249–255, 2007.

- [4] K. VanLehn, "The relative effectiveness of human tutoring, intelligent tutoring systems, and other tutoring systems," *Educational Psychologist*, vol. 46, no. 4, pp. 197–221, 2011.
- [5] Y. Leviathan, D. Valevski, M. Kalman, D. Lumen, E. Segalis, E. Molad, S. Pasternak, V. Natchu, V. Nygaard, V. Srinivasan, J. Manyika, and Y. Matias, "Generative UI: LLMs are effective UI generators," arXiv:2604.09577, 2026.
- [6] S. Tu, Y. Li, K. Chen, S. Zhang, J. Yu, D. Zhang-Li, L. Hou, J. Li, Y. Zhang, H. Liu *et al.*, "MAIC-UI: Making interactive courseware with generative UI," arXiv:2604.25806, 2026.
- [7] S. Kalyuga, P. Ayres, P. Chandler, and J. Sweller, "The expertise reversal effect," *Educational Psychologist*, vol. 38, no. 1, pp. 23–31, 2003.
- [8] O. Chen, S. Kalyuga, and J. Sweller, "The expertise reversal effect is a variant of the more general element interactivity effect," *Educational Psychology Review*, vol. 29, no. 2, pp. 393–405, 2017.
- [9] A. Renkl and R. K. Atkinson, "Structuring the transition from example study to problem solving in cognitive skill acquisition: A cognitive load perspective," *Educational Psychologist*, vol. 38, no. 1, pp. 15–22, 2003.
- [10] R. J. C. M. Salden, V. Aleven, R. Schwonke, and A. Renkl, "The expertise reversal effect and worked examples in tutored problem solving," *Instructional Science*, vol. 38, no. 3, pp. 289–307, 2010.
- [11] K. Z. Gajos, K. Everitt, D. S. Tan, M. Czerwinski, and D. S. Weld, "Predictability and accuracy in adaptive user interfaces," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI)*. ACM, 2008, pp. 1271–1274.
- [12] L. Findlater and J. McGrenere, "A comparison of static, adaptive, and adaptable menus," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI)*. ACM, 2004, pp. 89–96.
- [13] Z. A. Pardos and N. T. Heffernan, "Modeling individualization in a Bayesian networks implementation of knowledge tracing," in *User Modeling, Adaptation, and Personalization (UMAP 2010)*, ser. Lecture Notes in Computer Science. Springer, 2010, pp. 255–266.
- [14] M. V. Yudelson, K. R. Koedinger, and G. J. Gordon, "Individualized Bayesian knowledge tracing models," in *Artificial Intelligence in Education (AIED 2013)*, ser. Lecture Notes in Computer Science. Springer, 2013, pp. 171–180.
- [15] Y. Qiu, Y. Qi, H. Lu, Z. A. Pardos, and N. T. Heffernan, "Does time matter? modeling the effect of time with Bayesian knowledge tracing," in *Proceedings of the 4th International Conference on Educational Data Mining (EDM 2011)*, 2011, pp. 139–148.
- [16] M. Khajah, R. V. Lindsey, and M. C. Mozer, "How deep is knowledge tracing?" in *Proceedings of the 9th International Conference on Educational Data Mining (EDM 2016)*, 2016.
- [17] R. S. J. d. Baker, A. T. Corbett, and V. Aleven, "More accurate student modeling through contextual estimation of slip and guess probabilities in Bayesian knowledge tracing," in *Intelligent Tutoring Systems (ITS 2008)*, ser. Lecture Notes in Computer Science. Springer, 2008, pp. 406–415.
- [18] J. E. Beck and K.-m. Chang, "Identifiability: A fundamental problem of student modeling," in *User Modeling 2007*, ser. Lecture Notes in Computer Science. Springer, 2007, pp. 137–146.
- [19] B. van de Sande, "Properties of the Bayesian knowledge tracing model," *Journal of Educational Data Mining*, vol. 5, no. 2, pp. 1–10, 2013.
- [20] S. Doroudi and E. Brunskill, "The misidentified identifiability problem of Bayesian knowledge tracing," in *Proceedings of the 10th International Conference on Educational Data Mining (EDM 2017)*, 2017.
- [21] P. I. Pavlik, H. Cen, and K. R. Koedinger, "Performance factors analysis – a new alternative to knowledge tracing," in *Artificial Intelligence in Education (AIED 2009)*, ser. Frontiers in Artificial Intelligence and Applications. IOS Press, 2009, pp. 531–538.
- [22] Y. Gong, J. E. Beck, and N. T. Heffernan, "How to construct more accurate student models: Comparing and optimizing knowledge tracing and performance factor analysis," *International Journal of Artificial Intelligence in Education*, vol. 21, no. 1–2, pp. 27–46, 2011.
- [23] C. Piech, J. Spencer, J. Huang, S. Ganguli, M. Sahami, L. Guibas, and J. Sohl-Dickstein, "Deep knowledge tracing," arXiv:1506.05908, 2015.
- [24] R. Pelánek, "Bayesian knowledge tracing, logistic models, and beyond: An overview of learner modeling techniques," *User Modeling and User-Adapted Interaction*, vol. 27, no. 3–5, pp. 313–350, 2017.
- [25] M. C. Desmarais and R. S. J. d. Baker, "A review of recent advances in learner and skill modeling in intelligent learning environments," *User Modeling and User-Adapted Interaction*, vol. 22, no. 1–2, pp. 9–38, 2012.
- [26] P. Brusilovsky, "Adaptive hypermedia," *User Modeling and User-Adapted Interaction*, vol. 11, no. 1–2, pp. 87–110, 2001.
- [27] K. Z. Gajos, D. S. Weld, and J. O. Wobbrock, "Automatically generating personalized user interfaces with Supple," *Artificial Intelligence*, vol. 174, no. 12–13, pp. 910–950, 2010.
- [28] K. Z. Gajos, M. Czerwinski, D. S. Tan, and D. S. Weld, "Exploring the design space for adaptive graphical user interfaces," in *Proceedings of the Working Conference on Advanced Visual Interfaces (AVI)*. ACM, 2006, pp. 201–.
- [29] T. Lavie and J. Meyer, "Benefits and costs of adaptive user interfaces," *International Journal of Human-Computer Studies*, vol. 68, no. 8, pp. 508–524, 2010.
- [30] J. Sweller, "Cognitive load during problem solving: Effects on learning," *Cognitive Science*, vol. 12, no. 2, pp. 257–285, 1988.
- [31] J. Sweller, J. J. G. van Merriënboer, and F. G. W. C. Paas, "Cognitive architecture and instructional design," *Educational Psychology Review*, vol. 10, no. 3, pp. 251–296, 1998.
- [32] J. Sweller, J. J. G. van Merriënboer, and F. Paas, "Cognitive architecture and instructional design: 20 years later," *Educational Psychology Review*, vol. 31, no. 2, pp. 261–292, 2019.
- [33] J. Sweller, "Element interactivity and intrinsic, extraneous, and germane cognitive load," *Educational Psychology Review*, vol. 22, no. 2, pp. 123–138, 2010.
- [34] J. Sweller and G. A. Cooper, "The use of worked examples as a substitute for problem solving in learning algebra," *Cognition and Instruction*, vol. 2, no. 1, pp. 59–89, 1985.
- [35] E. R. Fyfe, N. M. McNeil, J. Y. Son, and R. L. Goldstone, "Concreteness fading in mathematics and science instruction: A systematic review," *Educational Psychology Review*, vol. 26, no. 1, pp. 9–25, 2014.
- [36] J. Vlassis, "The balance model: Hindrance or support for the solving of linear equations with one unknown," *Educational Studies in Mathematics*, vol. 49, no. 3, pp. 341–359, 2002.
- [37] M. Otten, M. van den Heuvel-Panhuizen, and M. Veldhuis, "The balance model for teaching linear equations: A systematic literature review," *International Journal of STEM Education*, vol. 6, no. 1, p. 30, 2019.
- [38] C. Kieran, "Concepts associated with the equality symbol," *Educational Studies in Mathematics*, vol. 12, no. 3, pp. 317–326, 1981.
- [39] N. Herscovics and L. Linchevski, "A cognitive gap between arithmetic and algebra," *Educational Studies in Mathematics*, vol. 27, no. 1, pp. 59–78, 1994.
- [40] M. W. Alibali, E. J. Knuth, S. Hattikudur, N. M. McNeil, and A. C. Stephens, "A longitudinal examination of middle school students' understanding of the equal sign and equivalent equations," *Mathematical Thinking and Learning*, vol. 9, no. 3, pp. 221–247, 2007.
- [41] J. L. Booth, C. Barbieri, F. Eyer, and E. J. Paré-Blagoev, "Persistent and pernicious errors in algebraic problem solving," *The Journal of Problem Solving*, vol. 7, no. 1, 2014.
- [42] K. J. Åström and T. Hägglund, *Advanced PID Control*. Research Triangle Park, NC: ISA – The Instrumentation, Systems, and Automation Society, 2006.
- [43] K. J. Åström and R. M. Murray, *Feedback Systems: An Introduction for Scientists and Engineers*. Princeton, NJ: Princeton University Press, 2008.
- [44] K. J. Åström and L. Rundqwist, "Integrator windup and how to avoid it," in *1989 American Control Conference*. IEEE, 1989, pp. 1693–1698.
- [45] J. L. Hellerstein, Y. Diao, S. Parekh, and D. M. Tilbury, *Feedback Control of Computing Systems*. Wiley, 2004.
- [46] R. Hunicke, "The case for dynamic difficulty adjustment in games," in *Proceedings of the 2005 ACM SIGCHI International Conference on Advances in Computer Entertainment Technology (ACE)*. ACM, 2005, pp. 429–433.
- [47] B. T. Willard and R. Louf, "Efficient guided generation for large language models," arXiv:2307.09702, 2023.
- [48] T. Scholak, N. Schucher, and D. Bahdanau, "PICARD: Parsing incrementally for constrained auto-regressive decoding from language models," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. ACL, 2021, pp. 9895–9901.
- [49] S. Geng, M. Josifoski, M. Peyrard, and R. West, "Grammar-constrained decoding for structured NLP tasks without finetuning," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. ACL, 2023, pp. 10932–10952.
- [50] G. Poesia, O. Polozov, V. Le, A. Tiwari, G. Soares, C. Meek, and S. Gulwani, "Synchronesh: Reliable code generation from pre-trained language models," arXiv:2201.11227, 2022.

- [51] S. Geng, H. Cooper, M. Moskal, S. Jenkins, J. Berman, N. Ranchin, R. West, E. Horvitz, and H. Nori, "JSONSchemaBench: A rigorous benchmark of structured outputs for language models," arXiv:2501.10868, 2025.
- [52] Z. R. Tam, C.-K. Wu, Y.-L. Tsai, C.-Y. Lin, H.-y. Lee, and Y.-N. Chen, "Let me speak freely? a study on the impact of format restrictions on large language model performance," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*. ACL, 2024, pp. 1218–1236.
- [53] J. Zhang, K. Vanacore, R. S. Baker, N. Ch, C. Mills, and O. Henkel, "How much mastery is enough mastery? the relationship between mastery in a lesson and the performance on the subsequent lesson," in *Proceedings of the 18th International Conference on Educational Data Mining (EDM 2025)*, 2025.
- [54] J. G. Ziegler and N. B. Nichols, "Optimum settings for automatic controllers," *Transactions of the ASME*, vol. 64, no. 8, pp. 759–765, 1942.
- [55] G. H. Cohen and G. A. Coon, "Theoretical consideration of retarded control," *Transactions of the ASME*, vol. 75, no. 5, pp. 827–834, 1953.
- [56] K. J. Åström and T. Hägglund, "Revisiting the Ziegler–Nichols step response method for PID control," *Journal of Process Control*, vol. 14, no. 6, pp. 635–650, 2004.
- [57] S. Skogestad, "Simple analytic rules for model reduction and PID controller tuning," *Journal of Process Control*, vol. 13, no. 4, pp. 291–309, 2003.
- [58] D. Graham and R. C. Lathrop, "The synthesis of "optimum" transient response: Criteria and standard forms," *Transactions of the American Institute of Electrical Engineers, Part II: Applications and Industry*, 1953.
- [59] Convai Innovations, "Laya: Open-weight decision model," Model card, <https://huggingface.co/convaiinnovations/laya>, 2026.
- [60] D. Kahneman, *Thinking, Fast and Slow*. New York: Farrar, Straus and Giroux, 2011.
- [61] F. Pezosa, J. L. Reutter, F. Suarez, M. Ugarte, and D. Vrgoč, "Foundations of JSON schema," in *Proceedings of the 25th International Conference on World Wide Web (WWW)*, 2016, pp. 263–273.
- [62] W. E. Hick, "On the rate of gain of information," *Quarterly Journal of Experimental Psychology*, vol. 4, no. 1, pp. 11–26, 1952.
- [63] N. Cowan, "The magical number 4 in short-term memory: A reconsideration of mental storage capacity," *Behavioral and Brain Sciences*, vol. 24, no. 1, pp. 87–114, 2001.
- [64] T. S. Tullis, "The formatting of alphanumeric displays: A review and analysis," *Human Factors*, vol. 25, no. 6, pp. 657–682, 1983.
- [65] J. I. Kiger, "The depth/breadth trade-off in the design of menu-driven user interfaces," *International Journal of Man-Machine Studies*, vol. 20, no. 2, pp. 201–213, 1984.
- [66] K. Larson and M. Czerwinski, "Web page design: Implications of memory, structure and scent for information retrieval," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI)*. ACM, 1998, pp. 25–32.
- [67] R. M. Dawes, "The robust beauty of improper linear models in decision making," *American Psychologist*, vol. 34, no. 7, pp. 571–582, 1979.
- [68] K. VanLehn, S. Ohlsson, and R. Nason, "Applications of simulated students: An exploration," *Journal of Artificial Intelligence in Education*, vol. 5, no. 2, pp. 135–175, 1994.
- [69] T. Käser and G. Alexandron, "Simulated learners in educational technology: A systematic literature review and a Turing-like test," *International Journal of Artificial Intelligence in Education*, vol. 34, no. 2, pp. 545–585, 2024.
- [70] S. E. Fancsal, T. Nixon, and S. Ritter, "Optimal and worst-case performance of mastery learning assessment with Bayesian knowledge tracing," in *Proceedings of the 6th International Conference on Educational Data Mining (EDM 2013)*, 2013.
- [71] S. Slater and R. S. Baker, "Degree of error in Bayesian knowledge tracing estimates from differences in sample sizes," *Behaviormetrika*, vol. 45, no. 2, pp. 475–493, 2018.
- [72] H. Levitt, "Transformed up-down methods in psychoacoustics," *The Journal of the Acoustical Society of America*, vol. 49, no. 2B, pp. 467–477, 1971.
- [73] K. M. Kelly, Y. Wang, T. Thompson, and N. T. Heffernan, "Defining mastery: Knowledge tracing versus n-consecutive correct responses," in *Proceedings of the 8th International Conference on Educational Data Mining (EDM 2015)*, 2015, pp. 630–631.
- [74] E. S. Page, "Continuous inspection schemes," *Biometrika*, vol. 41, no. 1/2, pp. 100–115, 1954.
- [75] M. Basseville and I. V. Nikiforov, *Detection of Abrupt Changes: Theory and Application*. Englewood Cliffs, NJ: Prentice Hall, 1993.
- [76] A. Galyardt and I. Goldin, "Predicting performance during tutoring with models of recent performance," arXiv:1501.02732, 2015.
- [77] J. Leppink, F. Paas, C. P. M. Van der Vleuten, T. Van Gog, and J. J. G. Van Merriënboer, "Development of an instrument for measuring different types of cognitive load," *Behavior Research Methods*, vol. 45, no. 4, pp. 1058–1072, 2013.
- [78] F. G. W. C. Paas, "Training strategies for attaining transfer of problem-solving skill in statistics: A cognitive-load approach," *Journal of Educational Psychology*, vol. 84, no. 4, pp. 429–434, 1992.
- [79] P. Ayres, "Using subjective measures to detect variations of intrinsic cognitive load within problems," *Learning and Instruction*, vol. 16, no. 5, pp. 389–400, 2006.
- [80] A. Badrinath, F. Wang, and Z. Pardos, "pyBKT: An accessible Python library of Bayesian knowledge tracing models," in *Proceedings of the 14th International Conference on Educational Data Mining (EDM 2021)*, 2021.
- [81] J. E. Beck and Y. Gong, "Wheel-spinning: Students who fail to master a skill," in *Artificial Intelligence in Education (AIED 2013)*, ser. Lecture Notes in Computer Science. Springer, 2013, pp. 431–440.